SQL PROJECT PRESENTATION

Healthcare Disease Analysis & Prediction

PROJECT TITLE	DATASET	BUSINESS DOMAIN	TOTAL RECORDS
Healthcare Disease Analysis & Prediction	disease prediction	Healthcare	10000

Q1 What is the disease prediction percentage?

PROBLEM STATEMENT Disease prevalence percentage

SQL QUERY

```
-- Q1.What is the disease prediction percentage?
select disease_prediction,count(*) as total_patient,
ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM cleaned), 2) AS percentage
from cleaned
group by disease_prediction ;
```

FINAL OUTPUT

	disease_prediction	total_patient	percentage
▶	No	910	75.83
	Yes	290	24.17

INSIGHT In this observation we can see that 75.83% of patients have no disease predicted and 24.17% of patients have a predicted disease.

Q2 Which age_group has the highest number of predicted disease cases?

PROBLEM STATEMENT Disease risk by age group.

SQL QUERY

```
select age_group,count(disease_prediction) as total_count
from cleaned
where disease_prediction = "Yes"
group by age_group
order by total_count desc;
```

FINAL OUTPUT

age_group	total_count
61-75	104
46-60	70
31-45	50
76+	44
18-30	22

INSIGHT The 61-75 age group had the highest risk of disease.

Q3 What is the disease rate for each sex?

PROBLEM STATEMENT Disease risk by gender

SQL QUERY

```
select sex,count(*) as total_patient,
ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM cleaned), 2) AS percentage
from cleaned
where disease_prediction = "Yes"
group by sex;
```

FINAL OUTPUT

	sex	total_patient	percentage
▶	Male	140	11.67
	Female	150	12.50

INSIGHT In this observation we see that 11.67% of males and 12.50% of females have a disease prediction.

Q4 Which age group has the highest average blood pressure?

PROBLEM STATEMENT Calculate average BP for each age group

SQL QUERY

```
select age_group, round(avg(blood_pressure),2) as average_BP
from cleaned
group by age_group
order by average_BP desc;
```

FINAL OUTPUT

age_group	average_BP
76+	126.33
61-75	124.84
18-30	124.25
31-45	123.35
46-60	123.08

INSIGHT Blood pressure is very high in patients aged 76+.

Q5 Which physical activity group has the highest average BMI?

PROBLEM STATEMENT Compare BMI across activity categories.

SQL QUERY

```
select physical_activity,avg(bmi) as average_bmi
from cleaned
group by physical_activity
order by average_bmi desc;
```

FINAL OUTPUT

	physical_activity	average_bmi
▶	High	26.44353348809812
	Moderate	26.259881533215566
	Low	26.071524521584532

INSIGHT The High, Moderate and Low groups all show nearly the same average, but looking at the minor difference, the High category has the maximum average compared to the other two categories.

Q6 Which prediction group has the highest average BMI, BP, glucose and cholesterol?

PROBLEM STATEMENT Compare multiple health indicators between Yes and No groups

SQL QUERY

```
select disease_prediction,round(avg(bmi),2) as average_bmi,round(avg(blood_pressure),2) as average_BP,
       round(avg(glucose_level),2) as average_glucose,round(avg(cholesterol),2) as average_cholesterol
from cleaned
group by disease_prediction
order by average_BP desc;
```

FINAL OUTPUT

disease_prediction	average_bmi	average_BP	average_glucose	average_cholesterol
Yes	26.85	127.33	117.05	210.2
No	26.05	122.96	107.98	201.18

INSIGHT The positive prediction group has a slightly higher average than the negative group in BMI, blood pressure, glucose level and cholesterol.

Q7 Which smoking group has the highest average blood pressure?

PROBLEM STATEMENT Compare BP between smokers and non-smokers.

SQL QUERY

```
select smoking,avg(blood_pressure) as average_BP
from cleaned
group by smoking
order by average_BP desc;
```

FINAL OUTPUT

	smoking	average_BP
▶	No	124.60260092176549
	Yes	122.1651852591631

INSIGHT In this observation we can see that non-smokers have a higher average blood pressure than smokers.

Q8 Which family-history group has the highest average glucose?

PROBLEM STATEMENT Compare glucose levels based on family history.

SQL QUERY

```
select family_history,round(avg(glucose_level),2) as average_glucose_level
from cleaned
group by family_history
order by average_glucose_level desc;
```

FINAL OUTPUT

	family_history	average_glucose_level
▶	No	110.99
	Yes	108.3

INSIGHT The highest average glucose level is in the family history category No, compared to the Yes category.

Q9 Which sex has the highest average cholesterol?

PROBLEM STATEMENT Compare cholesterol between sexes.

SQL QUERY

```
select sex,round(avg(cholesterol),2) as average_cholesterol
from cleaned
group by sex
order by average_cholesterol desc;
```

FINAL OUTPUT

	sex	average_cholesterol
▶	Male	204.43
	Female	202.47

INSIGHT Males have a higher cholesterol ratio than females.

Q10 Which age group has the highest percentage of smokers?

PROBLEM STATEMENT Calculate smoking percentage within each age group.

SQL QUERY

```
select age_group,smoking,ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM cleaned), 2) as percentage
from cleaned
where not smoking = "No"
group by age_group,smoking
order by percentage desc;
```

FINAL OUTPUT

	age_group	smoking	percentage
▶	31-45	Yes	6.67
	46-60	Yes	6.17
	61-75	Yes	5.42
	18-30	Yes	4.17
	76+	Yes	1.67

INSIGHT All age groups contribute to smoking, but the 31-45 and 46-60 age groups have the highest smoking ratio compared to other age groups.

Q11 What is the average sleep duration for Yes and No predictions?

PROBLEM STATEMENT Compare sleep hours between prediction groups.

SQL QUERY

```
select disease_prediction,round(avg(sleep_hours),2) as average_sleep_time
from cleaned
group by disease_prediction;
```

FINAL OUTPUT

	disease_prediction	average_sleep_time
▶	No	7.02
	Yes	6.86

INSIGHT In this observation we can see that the No prediction category has a higher average sleep time than the Yes prediction category.

Q12 Which age group has the highest average heart rate?

PROBLEM STATEMENT Compare heart rate across age groups.

SQL QUERY

```
select age_group,round(avg(heart_rate),2) as average_rate
from cleaned
group by age_group
order by average_rate desc;
```

FINAL OUTPUT

age_group	average_rate
76+	77.29
46-60	76.27
31-45	75.63
61-75	75.36
18-30	75.3

INSIGHT All age groups have an average heart rate of around 75, but looking at the minor difference, the 76+ age group has the highest average compared to the other age groups.

Q13 Which age-group + sex combination has the highest Yes-prediction rate?

PROBLEM STATEMENT Multi-factor demographic analysis.

SQL QUERY

```
select age_group,sex,count(*) as total_patient
from cleaned
where disease_prediction = "Yes"
group by age_group,sex
order by total_patient desc;
```

FINAL OUTPUT

age_group	sex	total_patient
61-75	Female	53
61-75	Male	51
46-60	Female	37
46-60	Male	33
31-45	Female	27
76+	Female	24
31-45	Male	23
76+	Male	20
18-30	Male	13
18-30	Female	9

INSIGHT The 61-75 age group, both male and female, has the highest occurrence of positive predictions.

Q14 Which patients have the highest BMI, glucose, cholesterol and blood pressure?

PROBLEM STATEMENT Extreme-value analysis

SQL QUERY

```
select patient_id,round(bmi,2)as bmi,glucose_level,cholesterol,blood_pressure
from cleaned
where bmi = (SELECT MAX(bmi) FROM cleaned)
   OR glucose_level = (SELECT MAX(glucose_level) FROM cleaned)
   OR cholesterol = (SELECT MAX(cholesterol) FROM cleaned)
   OR blood_pressure = (SELECT MAX(blood_pressure) FROM cleaned);
```

FINAL OUTPUT

patient_id	bmi	glucose_level	cholesterol	blood_pressure
P10001	39.21	104	134	135
P10056	29.6	187.5	114	132
P10088	26.7	90	303	122
P10107	39.21	97	219	119
P10118	26.4	117	303	117
P10200	28.7	187.5	214	145
P10231	25.9	102	303	139
P10359	29.4	187.5	169	116
P10424	39.21	160	126	155
P10434	26.3	131	303	85
P10437	27.4	93	110	168
P10501	39.21	65	230	141
P10516	31.4	97	188	168
P10528	29.3	187.5	220	150
P10655	22.3	101	303	134
P10661	21.1	65	303	104

INSIGHT The patients above have the highest BMI, glucose, cholesterol and blood pressure, as shown in the output.

Q15 Which combination of age group + sex + smoking + physical activity + family history has the highest Yes-prediction rate?

SQL QUERY

```sql
SELECT age_group,sex,smoking,physical_activity,family_history,COUNT(*) AS total_patients,
    SUM(CASE WHEN disease_prediction = 'Yes' THEN 1 ELSE 0 END) AS yes_predictions,
    ROUND(SUM(CASE WHEN disease_prediction = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),2) AS yes_prediction_rate
FROM cleaned
GROUP BY age_group,sex,smoking,physical_activity,family_history
ORDER BY yes_prediction_rate DESC
LIMIT 10;
```

FINAL OUTPUT

age_group	sex	smoking	physical_activity	family_history	total_patients	yes_predictions	yes_prediction_rate
76+	Female	Yes	Low	No	2	2	100.00
46-60	Female	Yes	Low	Yes	1	1	100.00
76+	Male	Yes	Moderate	Yes	1	1	100.00
61-75	Male	Yes	Low	Yes	1	1	100.00
18-30	Male	Yes	High	Yes	1	1	100.00
76+	Female	Yes	Low	Yes	1	1	100.00
61-75	Female	Yes	High	Yes	4	4	100.00
61-75	Male	No	High	Yes	5	4	80.00
76+	Male	No	Low	No	9	6	66.67
76+	Female	Yes	Moderate	Yes	3	2	66.67

INSIGHT The above combination has the highest Yes-prediction rate.



THANK YOU



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